

Introduction to Machine Learning

Prof. Dr. M. Elif Karslıgil
elif@yildiz.edu.tr

Course Information

- Instructor:
 - Prof. Dr. M. Elif Karslıgil
 - elif@yildiz.edu.tr
- Lectures:
 - Monday 14.00-17.00

Course Information

- Grading:
 - 2 mini projects (%15 each)
 - 1 final project (%20)
 - 1 midterm exam (%20)
 - Final exam (%30)

Course Information

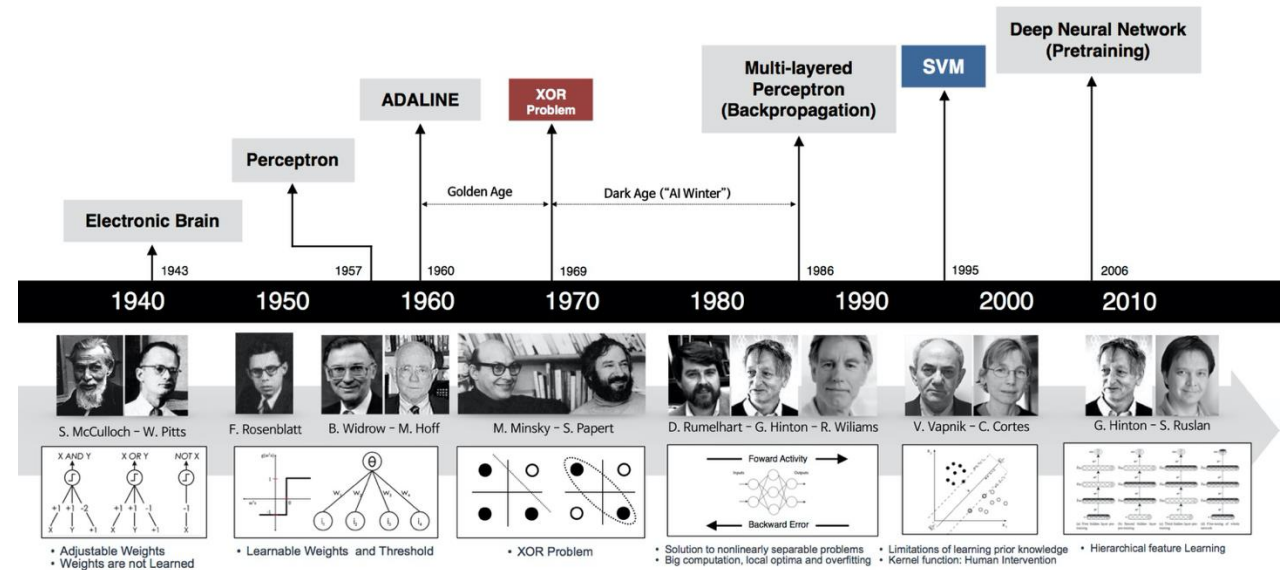
- Text Books: Not required, but for students who want to read more
 - Introduction to Machine Learning - 2nd Edition by Ethem Alpaydın, 2010
 - Pattern Recognition and Machine Learning by Christopher Bishop, 2006
 - Deep Learning by Ian Goodfellow, Yoshua Bengio, and Aaron Courville, 2016
 - Machine Learning Yearning by Andrew Ng, 2019

Prerequisites

- Basic familiarity with statistics
- Basic knowledge about algorithm design, data structures, algorithm complexity
- Coding in **Python**

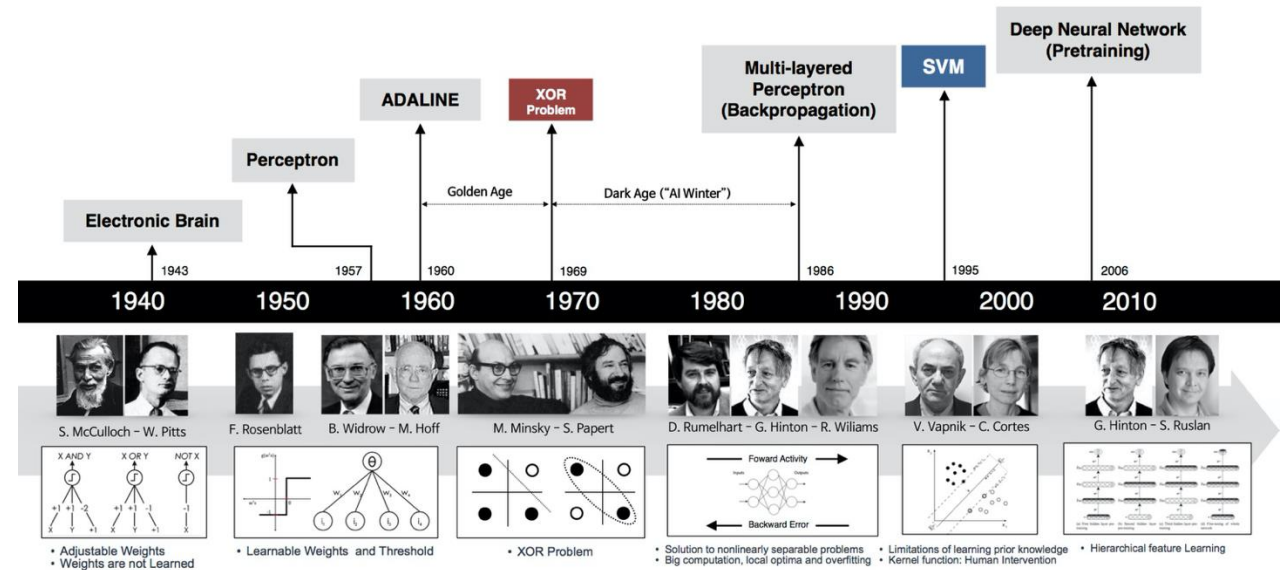
History of AI

- **1943:** Mc Culloch & Pitts: Boolean Circuit model of the brain
- **1950:** Alan Turing: Computing Machinery and Intelligence (Can machines think?)
- **1956:** Dartmouth meeting: The term Artificial Intelligence was coined by John McCarthy
- **1957:** Frank Resenblatt: perceptron, first NN model



History of AI

- **1966-1973:** Realization that many AI problems are intractable
- **1989:** Rumelhart, Hinton and Williams: popularized multilayered perceptron
- **2000:** Vladimir Vapnik et.al.: Support Vector Machine
- **2006:** Hinton et.al Deep Neural Network

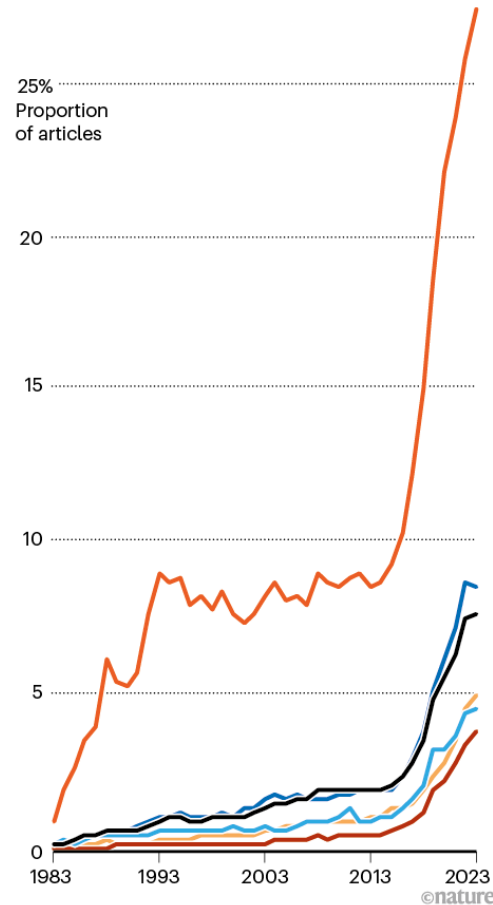


What 1,600 researchers think about AI?

AI ON THE RISE

The share of research papers with titles or abstracts that mention AI or machine-learning terms has risen to around 8%, analysis of the Scopus database suggests.

- Computer science
- Physical sciences
- Life sciences
- Social sciences
- Health and medicine
- Total

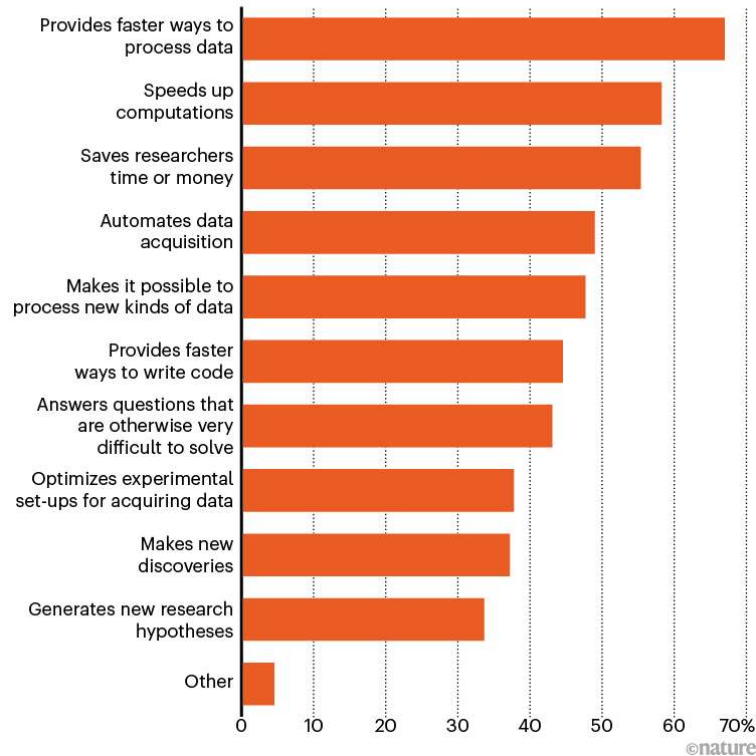


The share of research papers that mention AI terms has risen in every field over the past decade, according to an analysis for this article by *Nature*.

What 1,600 researchers think?

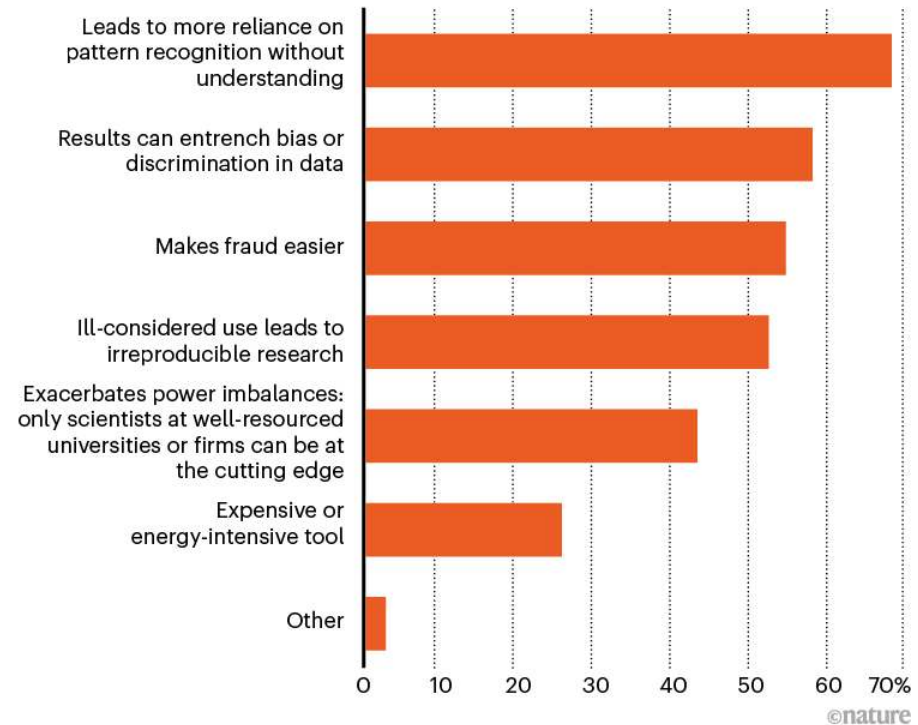
POSITIVE IMPACTS OF AI

Q: Considering machine-learning methods, what do you think are positive impacts of AI in research? (Choose all that apply.)



NEGATIVE IMPACTS OF AI

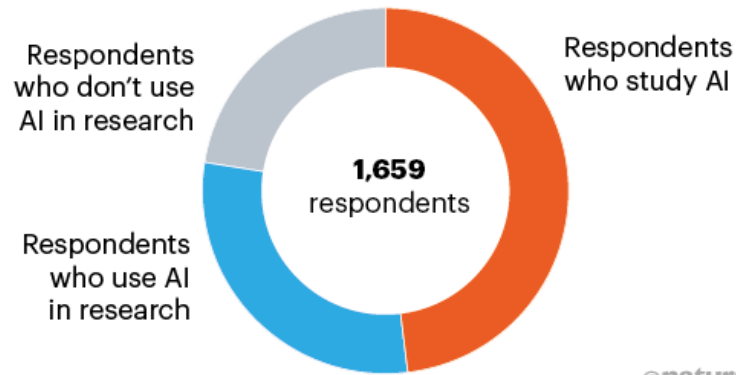
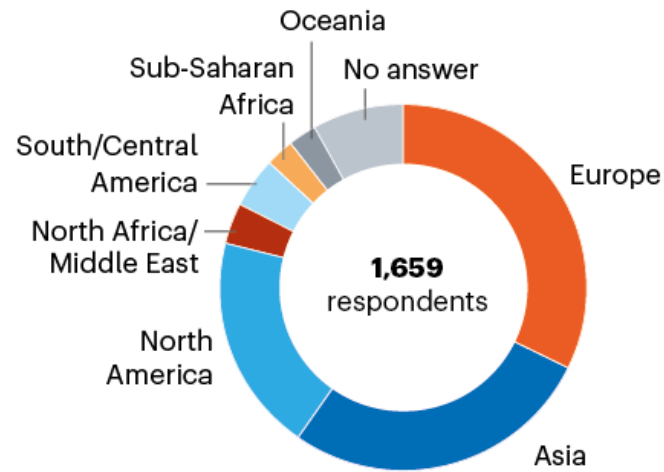
Q: Considering machine-learning methods, what do you think are negative impacts of AI in research? (Choose all that apply.)



AI and Research Survey Results

AI AND RESEARCH: SURVEY RESULTS

More than 1,600 respondents took *Nature's* survey on artificial intelligence (AI) and research. Just under half of respondents said they studied or developed AI themselves. Around one-third were from Europe, 28% from Asia and 20% from North America.



©nature

AI Anticipations

AI ANTICIPATIONS

Q: How useful do you think AI tools are for researchers in your field?

■ Essential ■ Very useful ■ Useful ■ Slightly useful ■ Not at all useful

Respondents who use AI in research



Respondents who don't use AI in research



Q: How useful do you think AI tools will become for researchers in your field in the next decade?

Respondents who use AI in research



Respondents who don't use AI in research



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What is Artificial Intelligence

- John McCarthy coined the phrase Artificial Intelligence in 1956
- The main goal of AI is to develop intelligent machines, that think, learn and act like humans.
- Some Subgoals
 - Perception
 - Reasoning
 - Search
 - Communication
 - Learning

What is Artificial Intelligence

- John McCarthy coined the phrase Artificial Intelligence in 1956
- The main goal of AI is to develop intelligent machines, especially intelligent computer programs.
- Some Subgoals
 - **Perception** : interpreting vision, sound, smell and touch
 - Reasoning
 - Search
 - Communication
 - Learning

What is Artificial Intelligence

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- Some Subgoals
 - Perception
 - **Reasoning:** getting logical conclusion and making predictions from available knowledge, facts, and beliefs
 - Search
 - Communication
 - Learning

What is Artificial Intelligence

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- Some Subgoals
 - Perception
 - Reasoning
 - **Search**: the process of navigating from a starting state to a goal state by transitioning through intermediate states
 - Communication
 - Learning

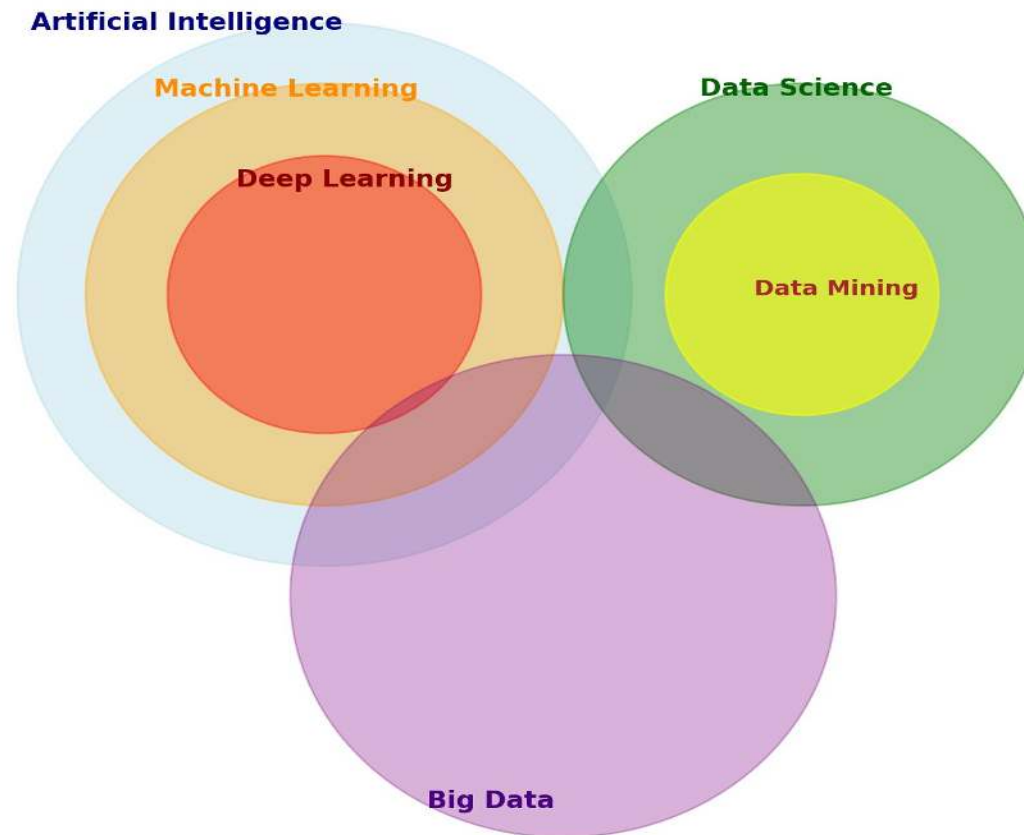
What is Artificial Intelligence

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- The main goal of AI is to develop intelligent machines, especially intelligent computer programs.
- Some Subgoals
 - Perception
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 - Search
 - **Communication:** providing human-computer interaction. Technologies such as machine translation of human languages, spoken dialogue systems like Siri
 - Learning

What is Artificial Intelligence

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 - Perception
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 - Communication
 - Learning: by MACHINE LEARNING

Artificial Intelligence and Related Fields



What is Machine Learning ?

- Machine Learning is the field of study that gives the computer **the ability to learn without being explicitly programmed** (Arthur Samuel -1959) .
- ML systems improve their learning over time, by feeding them data and information in the form of observations and real-world interactions

Traditional Programming



- Applications
 - Web Page coding
 - Bank ATM software
 - Tetris game
 - Stock control software

Traditional Programming

- How can we write a computer program?
 - we design an algorithm that encodes a set of rules that are useful to solve the problem
 - For some problems it is very difficult to specify those rules
 - Is there a cat in the image ?



What is Machine Learning ?

- Traditional Programming



- Machine Learning Programming



What is Machine Learning ?

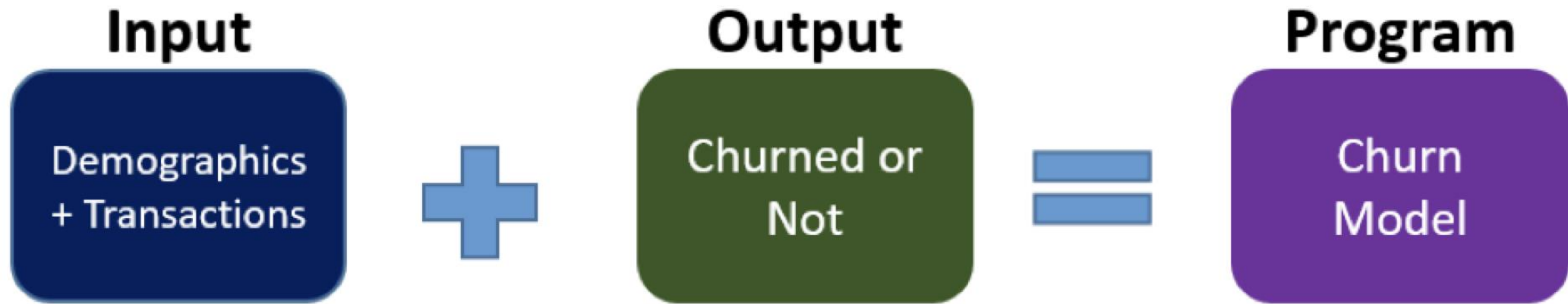
- Machine Learning Programming:



- Instead of writing a program, we collect examples that specify the correct output for a given input
- A machine learning algorithm then takes these examples and produces program that does the job
- If we do it right, the program works for new cases as well as the ones we trained it on.

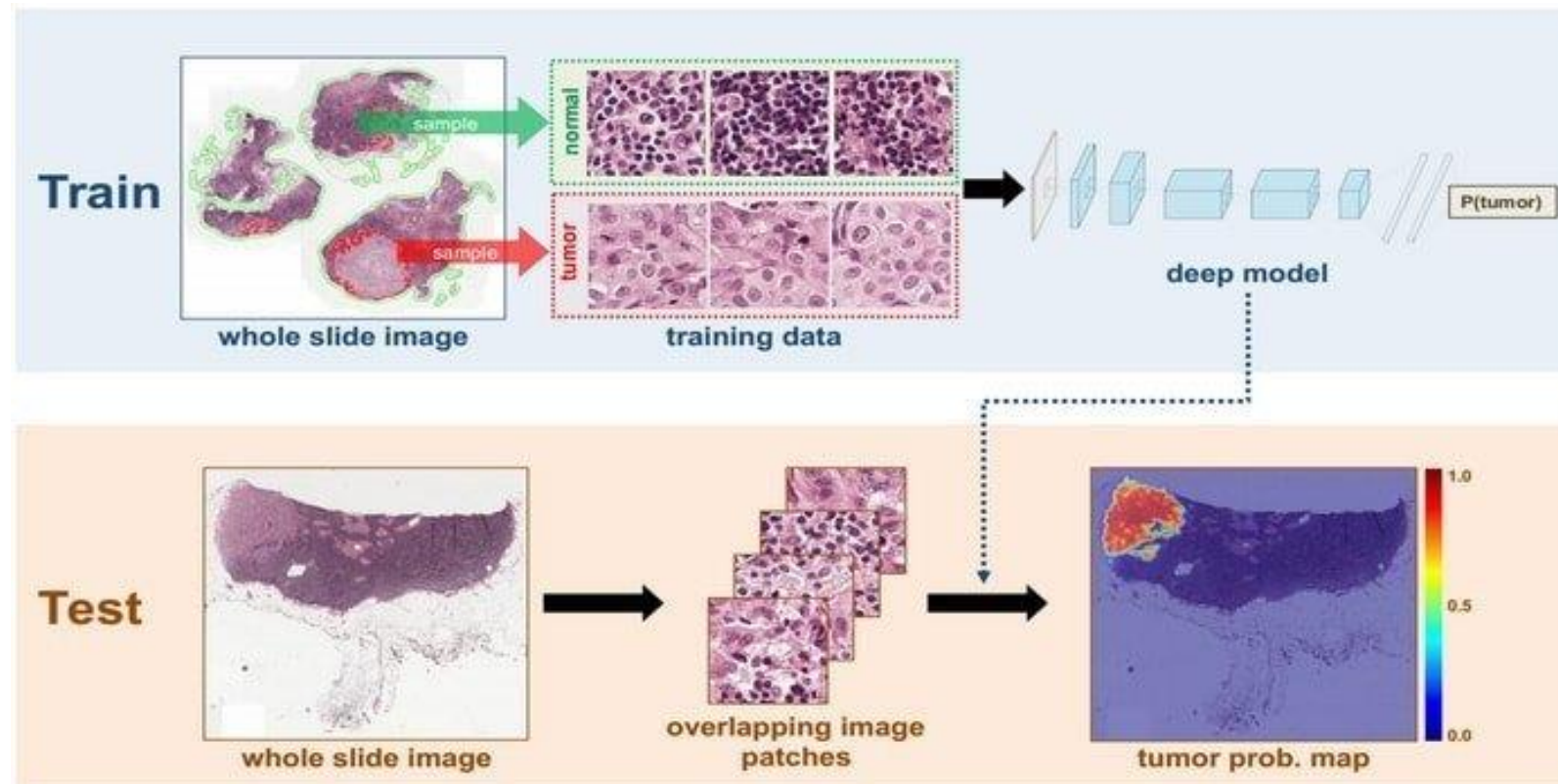
What is Machine Learning ?

- Example : Predict if a customer will churn or not



Machine Learning Applications

- Cancer Detection and Prediction



Machine Learning Applications

- Robotic Surgery



Machine Learning Applications

- Virtual Personal Assistants



SIRI



ALEXA



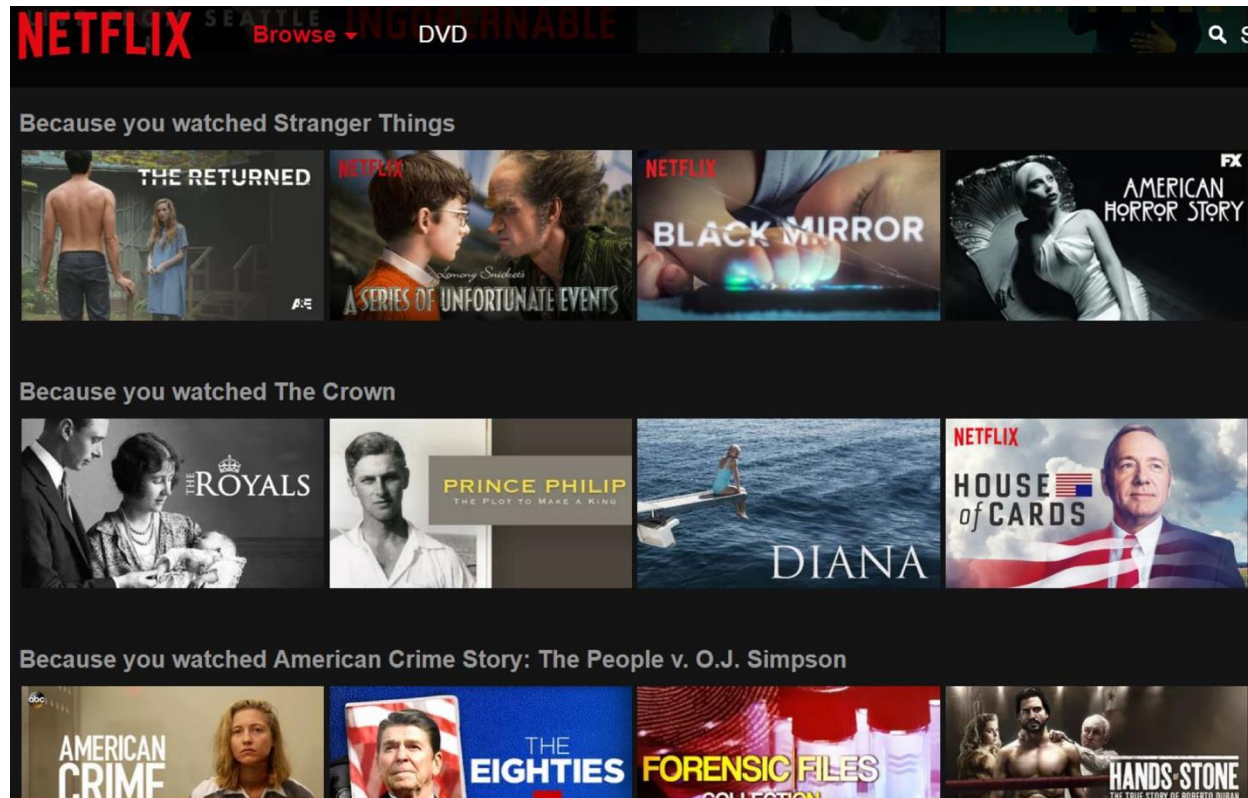
GOOGLE



CORTANA

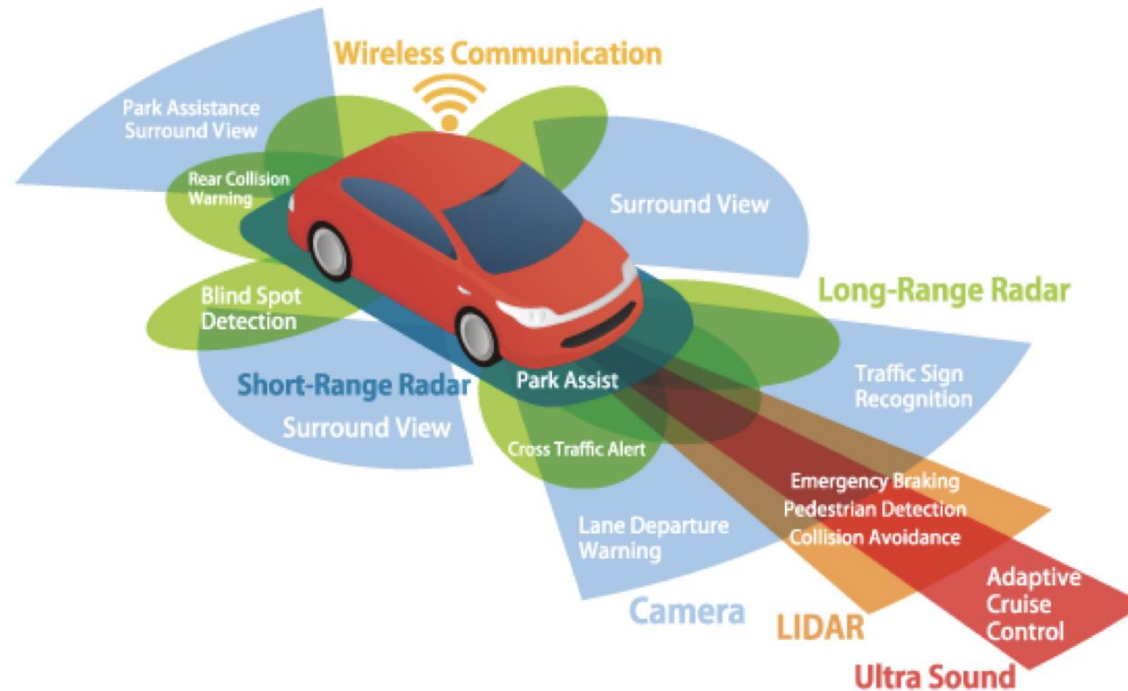
Machine Learning Applications

- Recommendation Systems



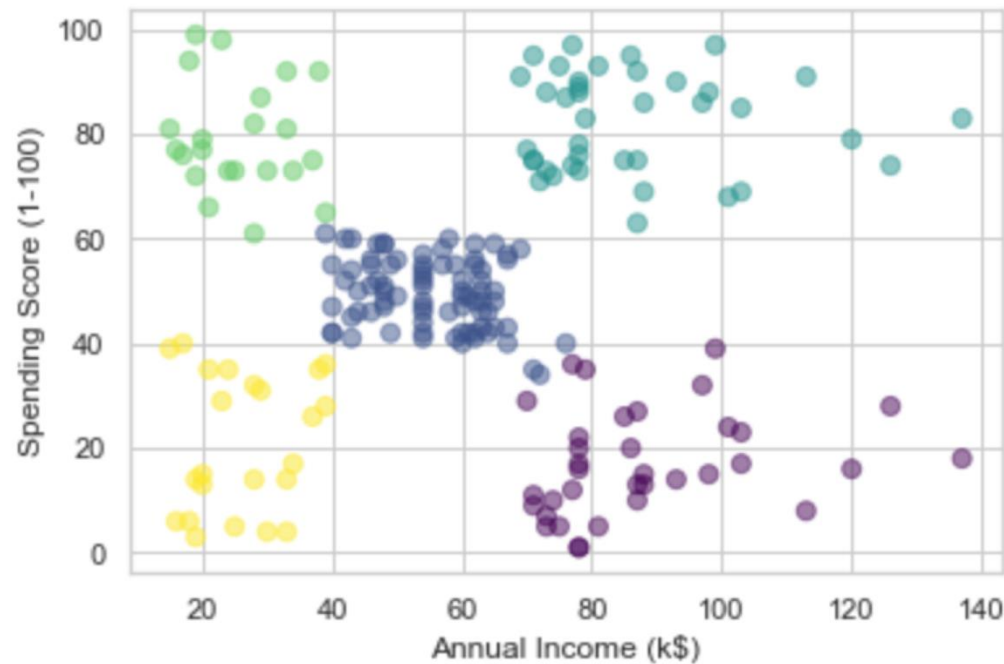
Machine Learning Applications

- Self Driving Vehicle



Machine Learning Applications

- Customer Segmentation



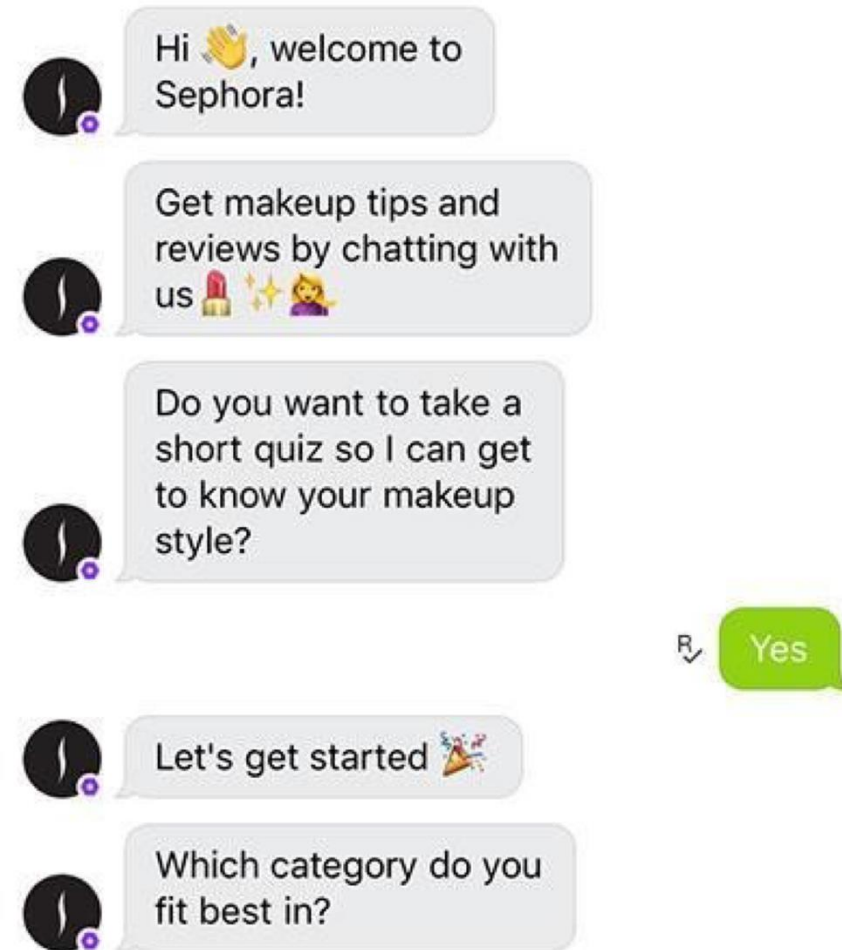
Machine Learning Applications

- Game Playing



Machine Learning Applications

- Automatic Chatbot



Machine Learning Applications

ChatGPT



Developer(s) [OpenAI](#)

Initial release November 30, 2022; 10 months ago

Stable release August 3, 2023; 59 days ago^[1]

Written in [Python](#)

Engine [GPT-3.5](#)
[GPT-4](#)

Platform [Cloud computing platforms](#)

Type [Chatbot](#)
[Large language model](#)
[Generative pre-trained transformer](#)

License [Proprietary](#)

Website chat.openai.com/chat [↗](#)

- **ChatGPT** : **Chat Generative Pre-trained Transformer**, is a [large language model](#)-based [chatbot](#) developed by [OpenAI](#) and launched on November 30, 2022,
- which enables users to refine and steer a conversation towards a desired length, format, style, level of detail, and language.

Why is Machine Learning Important?

- **Data in many domain is huge**
 - Thousands to billions of data samples
 - Impossible for human analysts to see patterns across so much data
- In some domains discovery and use of patterns must happen in real time, e.g. in streaming data

Types of Machine Learning

- Supervised Learning (predictive-predict the feature)
 - Build a predictive model from examples of data with known outcomes
 - Learn to **predict output** for a given input vector
- Unsupervised Learning (descriptive-understand the past)
 - Discover structure in data for which outcomes are not known

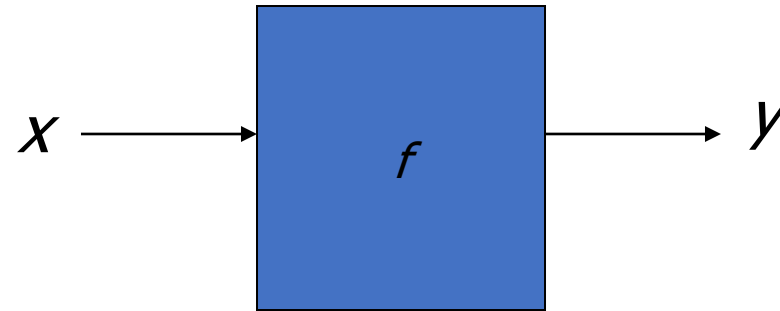
Supervised Learning

- Learn to predict output when given an input vector
- Samples : $D = \{d_1, d_2, \dots, d_n\}$

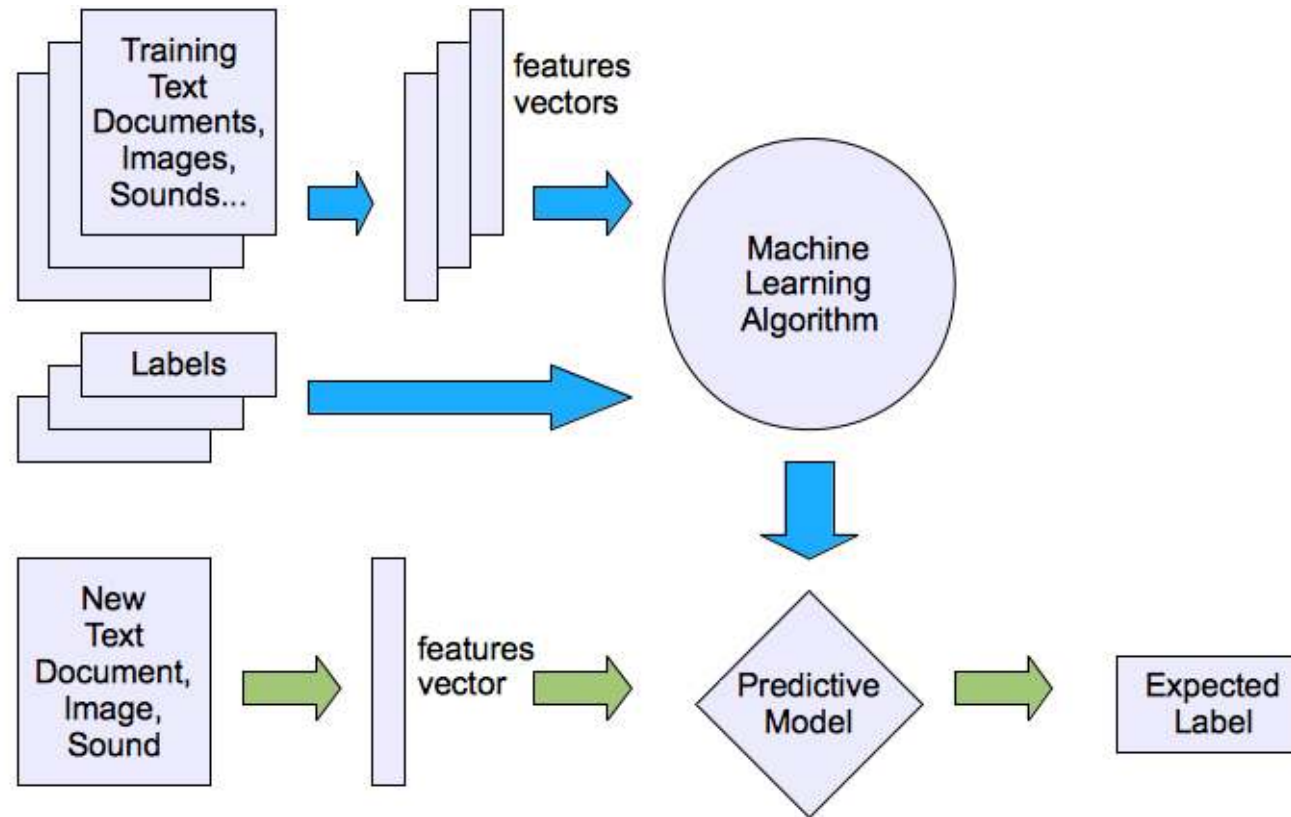
$$d_i = \langle \mathbf{x}_i, \mathbf{y}_i \rangle$$

$\mathbf{x}_i$: Input vector

$\mathbf{y}_i$: Desired Output



Supervised Learning

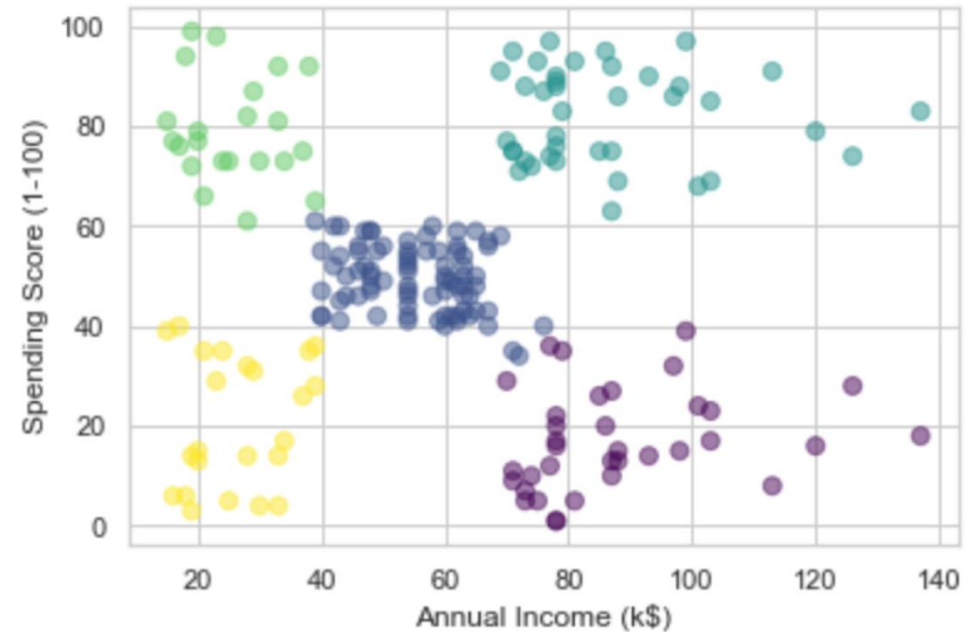


Unsupervised Learning

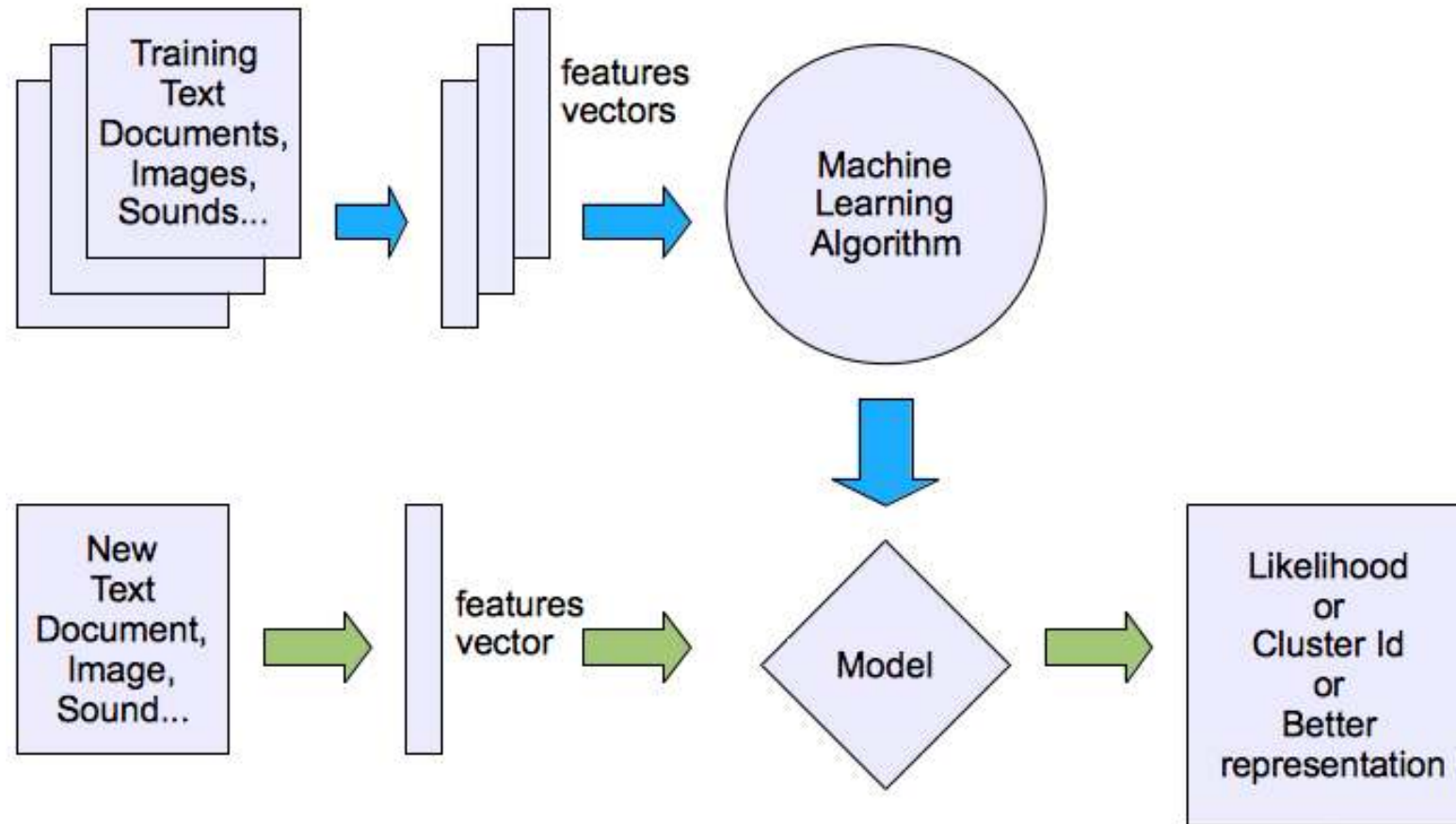
- Models are not supervised using training dataset
- Samples : $D = \{d_1, d_2, \dots, d_n\}$

$$d_i = \langle \mathbf{x}_i \rangle$$

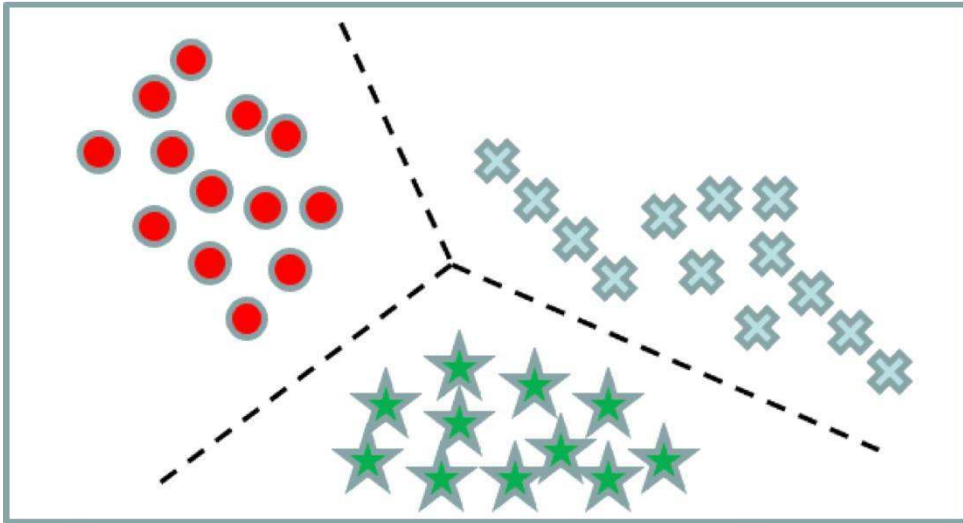
$\mathbf{x}_i$: Input vector



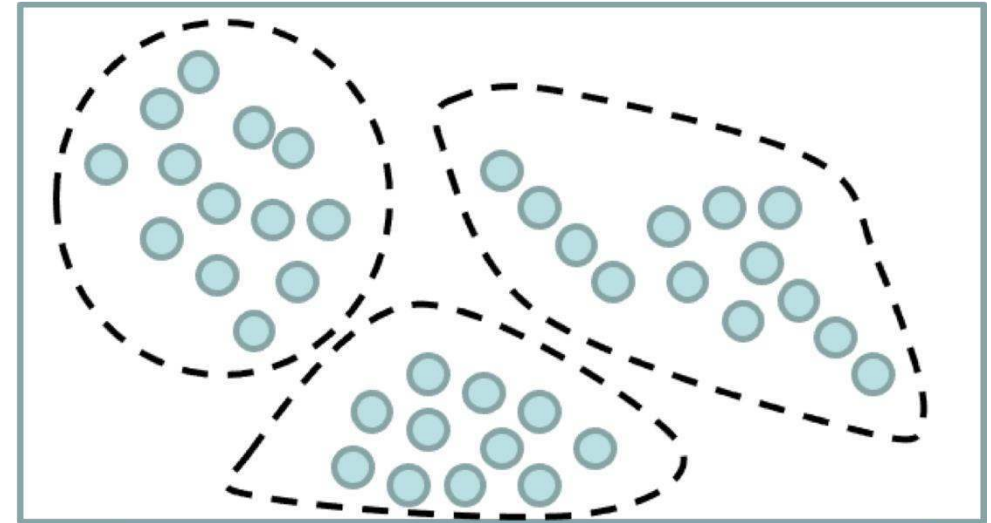
Unsupervised Learning



Types of Machine Learning

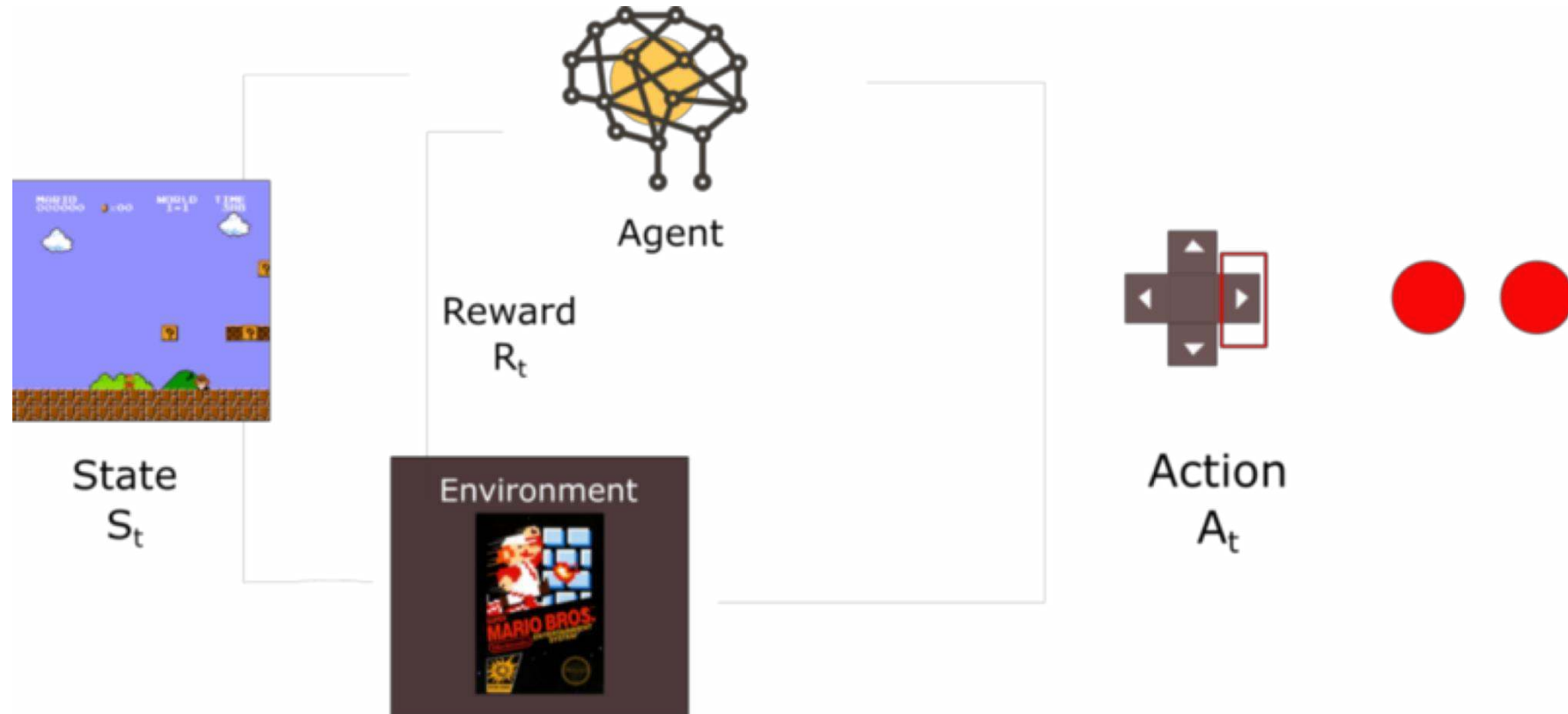


Eğitmenli Öğrenme

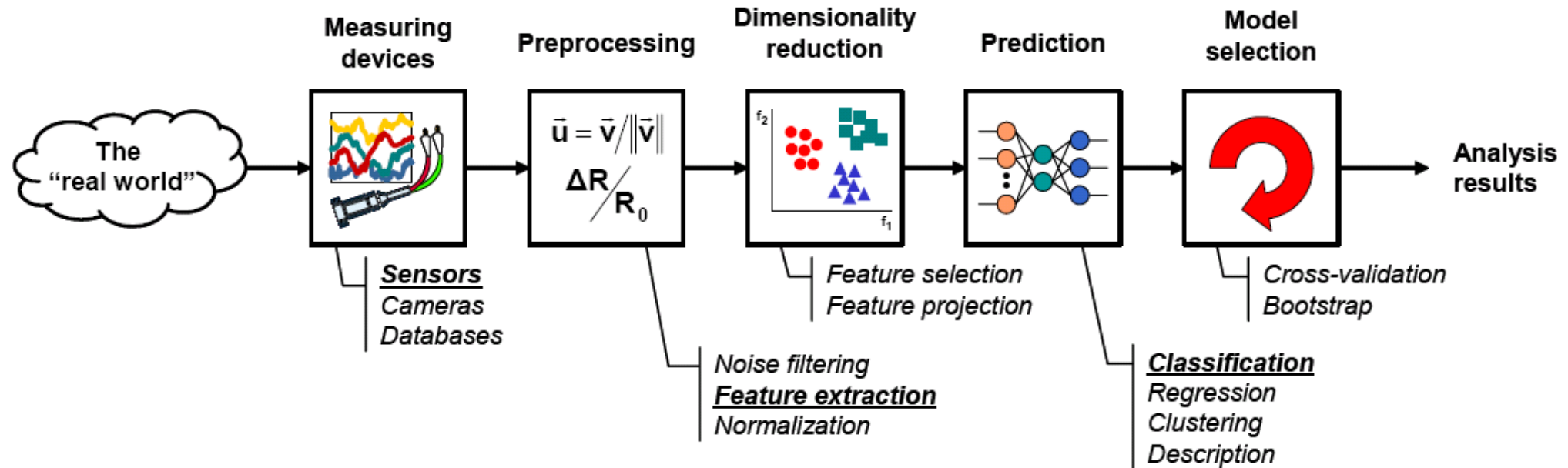


Eğitmensiz Öğrenme

Reinforcement Learning



Stages of a supervised machine learning system



Example Problem:

Credit Card Fraud Detection

Steps:

1. Determining Learning Approach
2. Data Set Preperation
3. Assumptions
4. Feature Selection-Extraction
5. Model Selection
6. Evaluation

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Example Problem:

Credit Card Fraud Detection

- Determining Learning Approach:
 - Can the data be labeled?
 - YES (Fraud, Normal)
 - Is data labeling hard? Is it time consuming?
 - NO

Example Problem:

Credit Card Fraud Detection

Steps:

1. Determining Learning Approach
2. Data Set Preperation
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6. Evaluation

Example Problem:

Credit Card Fraud Detection

- Data Set Preperation
 - Establish data collection mechanism
 - Acquire and integrate data
 - Check your data quality
 - Handle missing data
 - Preprocess (Format, Clean, Normalize etc.)
 - Transform the data if necessary
 - Split into training and evaluation sets

Example Problem:

Credit Card Fraud Detection

- Data Set Preparation
 - How much data is enough?
 - From which sources should i collect data?
 - Who could label the data?
 - How common are label errors?
 - Do we need to create new features ?
 - Can synthetic data be used?
 - How could outliers be handled ?

Example Problem:

Credit Card Fraud Detection

Steps:

1. Determining Learning Approach
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Example Problem:

Credit Card Fraud Detection

- Assumptions
 - Only make prediction for online shopping
 - Do not profile a new customer
 - Make prediction for above 200 TL

Example Problem:

Credit Card Fraud Detection

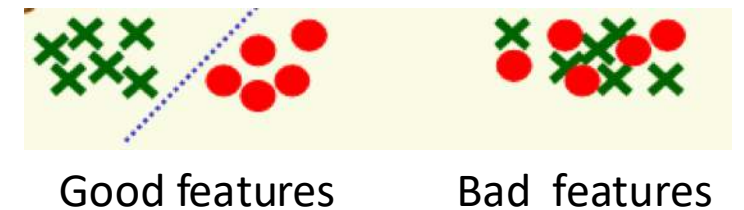
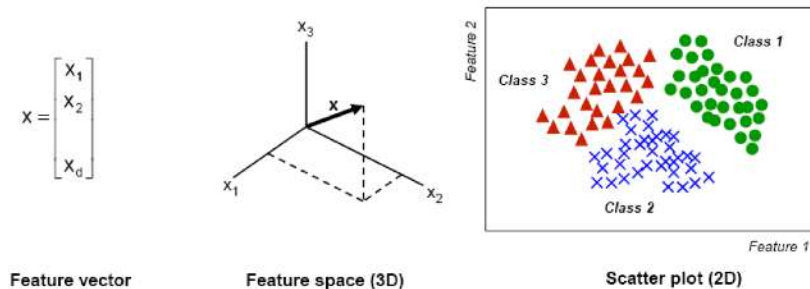
Steps:

1. Determining Learning Approach
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Example Problem:

Credit Card Fraud Detection

- Feature Selection-Extraction
 - Features are input to the ML system
 - Reducing features improves performance of the model
 - Reducing features reduce computational cost



Example Problem:

Credit Card Fraud Detection

Steps:

1. Determining Learning Approach
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3. Assumptions
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Example Problem:

Credit Card Fraud Detection

- Model Selection: We have a set of models $M=\{M1, M2..Md\}$
 - Which model generalizes the best?
 - Model selection
 - Hyperparameter selection
 - Evaluate complexity, maintainability, and available resources of a model .

Example Problem:

Credit Card Fraud Detection

Steps:

1. Determining Learning Approach
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Example Problem:

Credit Card Fraud Detection

- Evaluation:
- How to evaluate the performance of a model?
 - Training and validation sets are used to simulate unseen data
 - First the common problems should be identified, then test scenarios are prepared
 - It is important that the test data is not used *in any way* to create the classifier